First-Order Logic: Masterclass & Knowledge Check

COM759 Week 4 | The Architecture of Reason



Gemini Notebook

The Logic Taxonomy Matrix

Logic System	Ontological Commitment	Epistemological Commitment	Decidability
Propositional Logic	World contains facts	True/False/Unknown	Fully Decidable
First-Order Logic	World contains objects, $\forall x\, P(x)$ relations, $f(x,y)$ functions	True/False/Unknown	Semi-Decidable (RE)
Higher-Order Logic	Quantifies over predicates and functions $\exists F\, F(x)$ $G(f)$	True/False/Unknown	Undecidable $\neg$RE

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Question: Predicates vs. Propositions

Suppose $Q(x, y)$ is $x + y > 4$, where the universe of discourse is all integer pairs. Which of the following correctly classifies $Q(1000, 2)$ and $Q(x, 2)$?

A) Both are propositions.

B) $Q(1000, 2)$ is a proposition (True); $Q(x, 2)$ is a predicate.

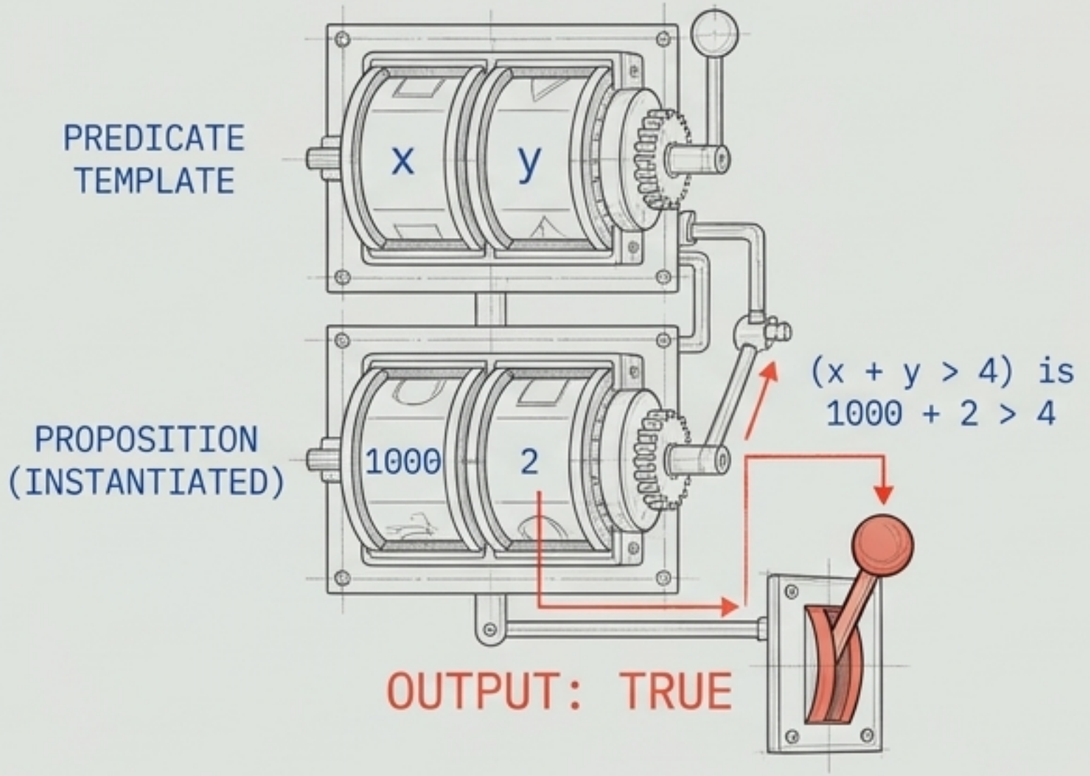
C) $Q(1000, 2)$ is a predicate; $Q(x, 2)$ is a proposition.

D) Both are predicates.

Correct Answer: B

Rule:

A predicate is a template with open variables (no truth value). When constants lock in, it becomes a proposition (outputs True or False).



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Analyze the following strings:

1) `f(g(x, y), h(x, y, z), y)` and 2) `f(a, b) = c`. How are they classified in FOL syntax?

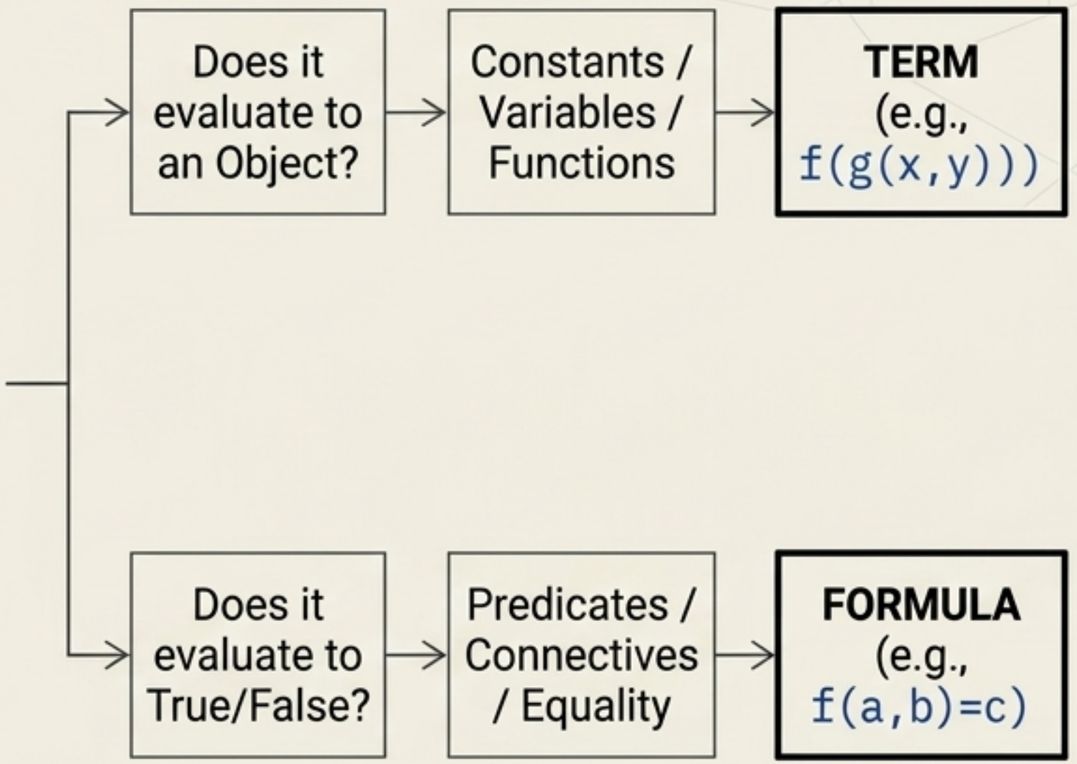
A) 1 is a formula, 2 is a term.

B) Both are formulas.

C) 1 is a function term, 2 is a formula.

D) Both are function terms.

Correct Answer: C



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Which of the following is a Well-Formed Formula (WFF) containing NO free variables?

A) $\exists x\ P(y)$

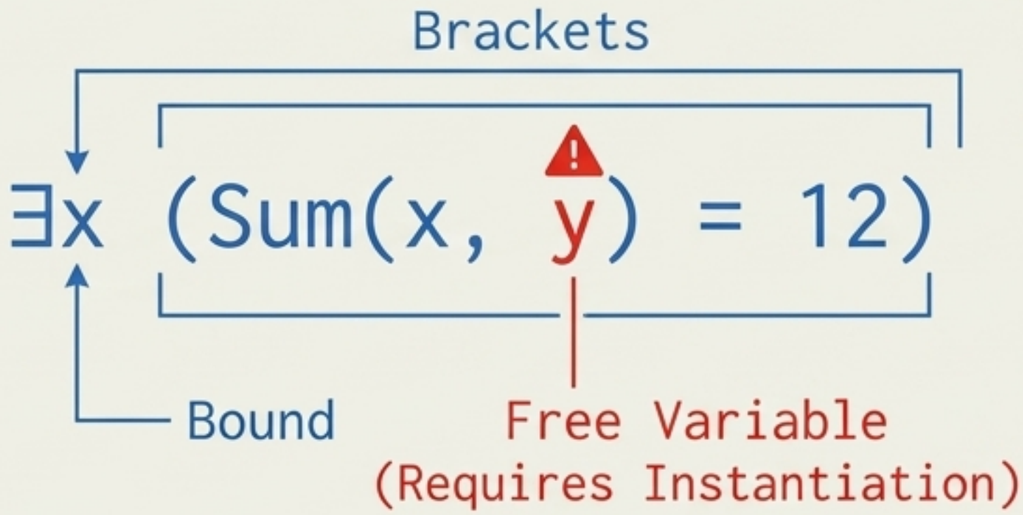
B) $P(x) \rightarrow \forall x\ Q(x)$

C) $\exists x\ (Sum(x, 3) = 12)$

D) $\exists x\ (Sum(x, y) = 12)$

Correct Answer: C

Rule: A WFF is a sentence containing no free variables; everything must be bound by a quantifier.



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Which is the correct FOL representation for:
There is an FBK-working smart person?

A) $\exists x\ (\text{WorksAt}(\text{FBK}, x) \rightarrow \text{Smart}(x))$

B) $\forall x\ (\text{WorksAt}(\text{FBK}, x) \rightarrow \text{Smart}(x))$

C) $\exists x\ (\text{WorksAt}(\text{FBK}, x) \land \text{Smart}(x))$

D) $\forall x\ (\text{WorksAt}(\text{FBK}, x) \land \text{Smart}(x))$

Correct Answer: C

Danger: Using → with ∃ means: There is a person so that if they work at FBK, they are smart.

This evaluates to True just by finding anyone who doesn't work at FBK!

QUANTIFIERS	CONNECTIVES
$\forall$ —— →	Universal Rules
$\exists$ —— $\land$	Existential Lists

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Let the universe of discourse be pairs of real numbers. Evaluate the truth of these two nested quantifier statements:

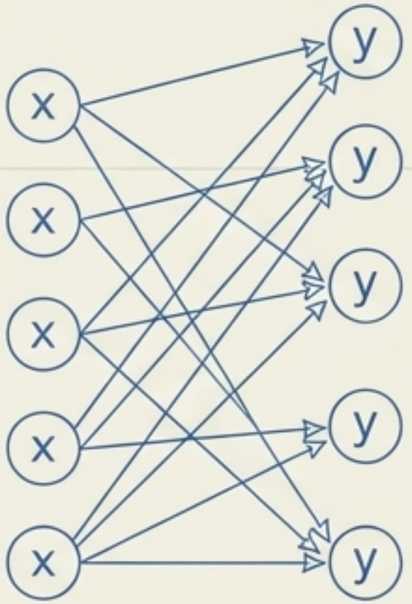
1) $\forall x \exists y (y > x)$ and
2) $\exists y \forall x (y > x)$.

A) Both 1 and 2 are True.
B) 1 is True, 2 is False.
C) 1 is False, 2 is True.
D) Both 1 and 2 are False.

Correct Answer: B

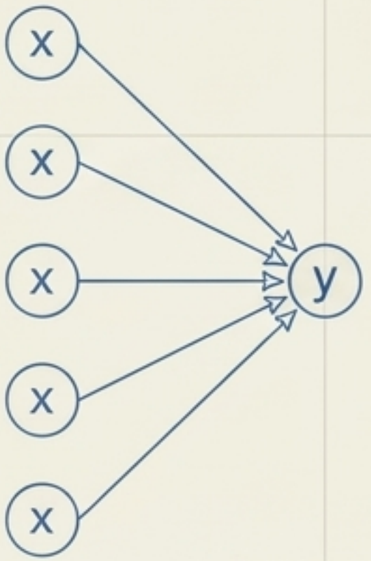
Rule: Switching universals and existentials changes meaning entirely.

$\forall x \exists y$



For every number, there is a bigger one (True)

$\exists y \forall x$



There is one supreme number bigger than all others (False)

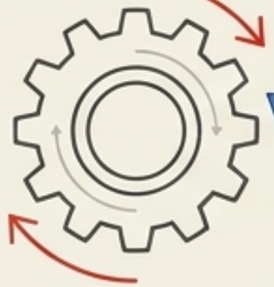
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Let D(x) be x majors in computer science. Which statement is logically equivalent to ¬∀x D(x) (Not every student majors in CS)?

A) ∃x ¬D(x)
B) ∀x ¬D(x)
C) ¬∃x D(x)
D) ∃x D(x)

Correct Answer: A

1) ¬[∀x P(x)]

2) ¬[]

3) ∃x ¬P(x)

There is a student who does NOT major in CS.

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What is the correct formalization for:
At least 2 students attend the KR course?

A) $\exists x1\ \exists x2\ (attend(x1,KR) \land attend(x2,KR))$

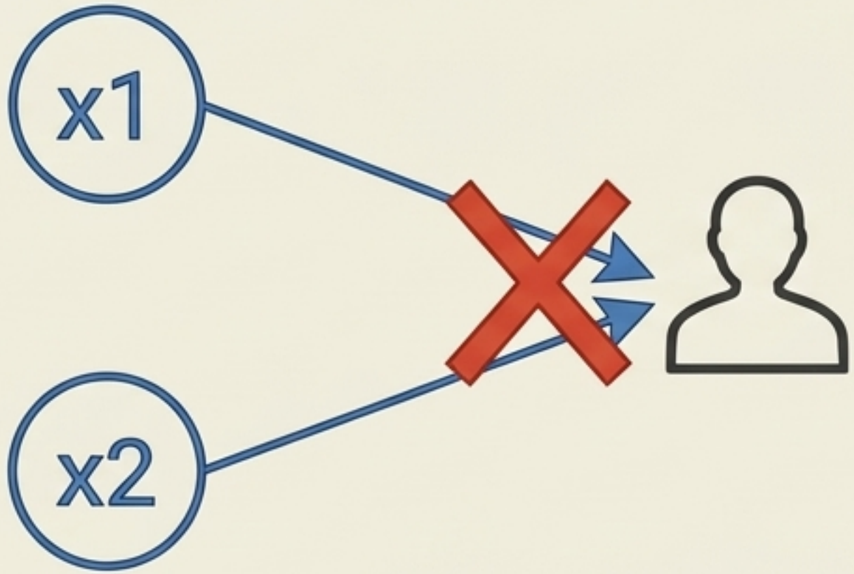
B) $\exists x1\ \exists x2\ (attend(x1,KR) \land attend(x2,KR) \land x1 \neq x2)$

C) $\forall x1\ \forall x2\ (attend(x1,KR) \rightarrow attend(x2,KR))$

D) $\exists x1\ \exists x2\ (attend(x1,KR) \lor attend(x2,KR) \land x1 \neq x2)$

Correct Answer: B

Rule: Without the explicit x1 ≠ x2 constraint, FOL allows multiple variables to point to the exact same object in the domain.



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Apply the **Existential Instantiation** rule to the premise $\exists x\ eats(Ziggy, x)$. Which of the following is the correct deduction?

A) $eats(Ziggy, IceCream)$, where $IceCream$ is an existing known constant.

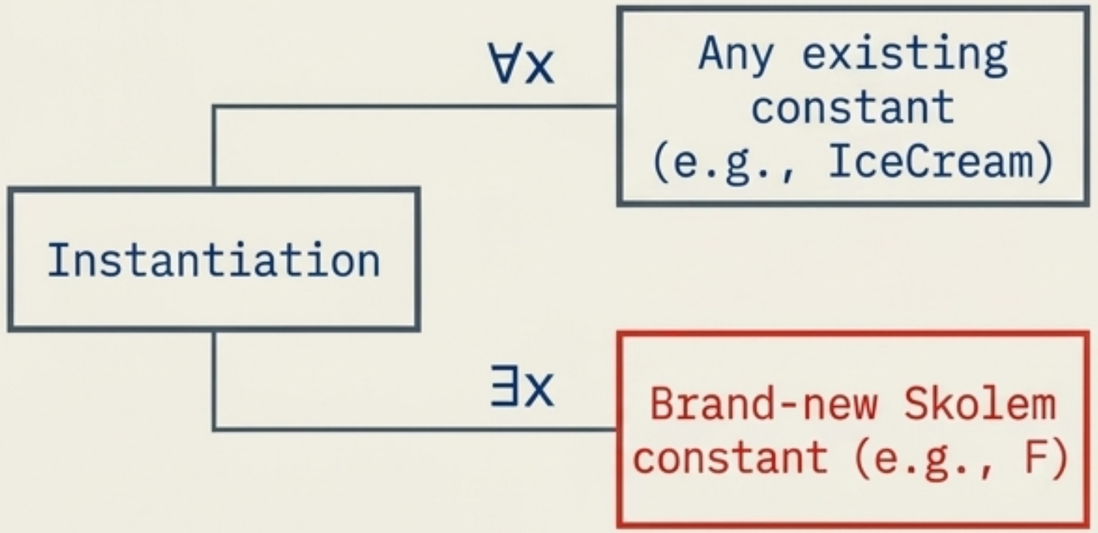
B) $eats(Ziggy, y)$, where y is a universally quantified variable.

C) $eats(Ziggy, F)$, where F is a brand-new Skolem constant not occurring anywhere else in the KB.

D) $\forall x\ eats(Ziggy, x)$

Correct Answer: C

Rule: We replace the existential variable with a new constant to reason about the unknown object without manipulating the quantifier continuously.



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What is the Most General Unifier (MGU) of the atomic formulas $P(a, x)$ and $P(y, g(y))$?

A) $\{a/y, b/x\}$

B) Fail

C) $\{a/x, g(a)/y\}$

D) $\{a/y, g(a)/x\}$

Correct Answer: D

Step 1: Match first arguments: a unifies with y. Substitution: $\{a/y\}$.

Step 2: Match second arguments: x unifies with $g(y)$. Apply existing substitution $(y=a)$, so x unifies with $g(a)$. Substitution: $\{g(a)/x\}$.

Term 1	Term 2	MGU
$P(a, x)$	$P(y, g(y))$	$\{\}$
a	y	$\{a/y\}$
x	$g(y)$	$\{a/y, g(a)/x\}$

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Attempt to unify the atomic formulas `P(x, f(x))` and `P(y, y)`. What is the result?

A) `{y/x, f(x)/y}`

B) `{f(y)/x}`

C) `{f(x)/y, y/x}`

D) FAIL (occurs check)

Correct Answer: D

Rule: A variable cannot unify with a function term that contains that same variable.

	Term 1	Term 2	MGU
1	`P(x, f(x))`	`P(y, y)`	`{}`
2	`x`	`y`	`{y/x}`
3	`f(x)`	`y` → `f(y)`	**FAIL (occurs check)**

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In the semantics of FOL, an Interpretation function over a domain Δ does **ALL** of the following **EXCEPT:**

A) Assigns each `constant` to an `object` in Δ.

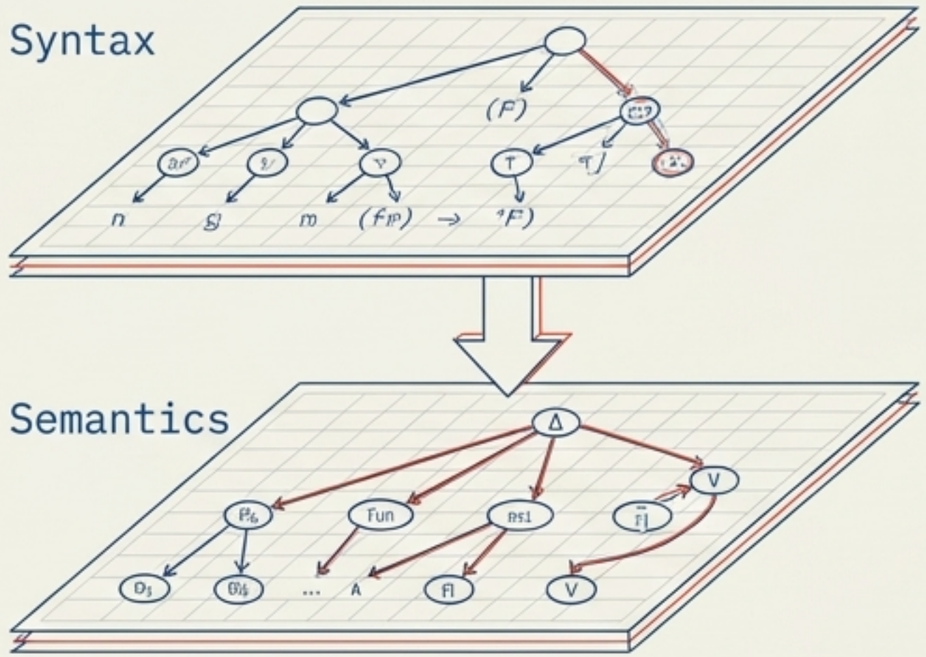
B) Maps each `n-ary predicate` to a `truth value`.

C) **Automatically proves that every sentence is** `universally valid`.

D) Defines each `n-ary function` as a mapping $\Delta^n \rightarrow \Delta$.

Correct Answer: C

Rule: An interpretation assigns meaning. Satisfiability and validity are logical consequences evaluated under these interpretations, not automatically generated by them.



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Which of the following statements about the decidability of FOL Knowledge Base entailment is true?

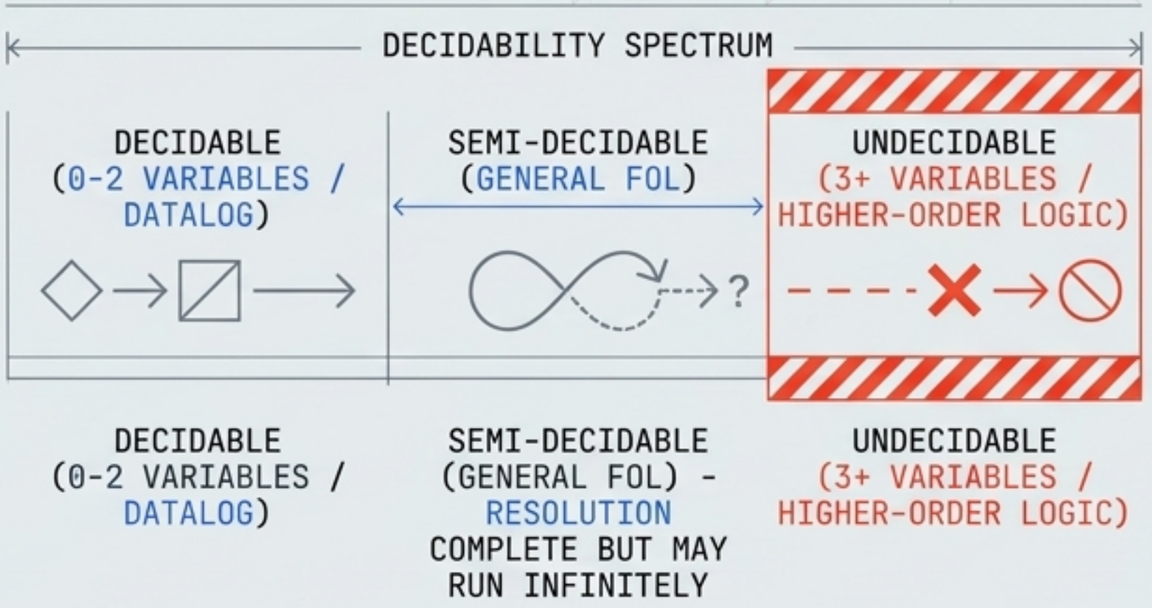
A) It is fully decidable for any number of variables.

B) It is semi-decidable in general, but decidable if every sentence has at most 2 variables.

C) It is entirely undecidable for any subset of FOL.

D) Entailment is resolved instantly using the Resolution algorithm.

Correct Answer: B



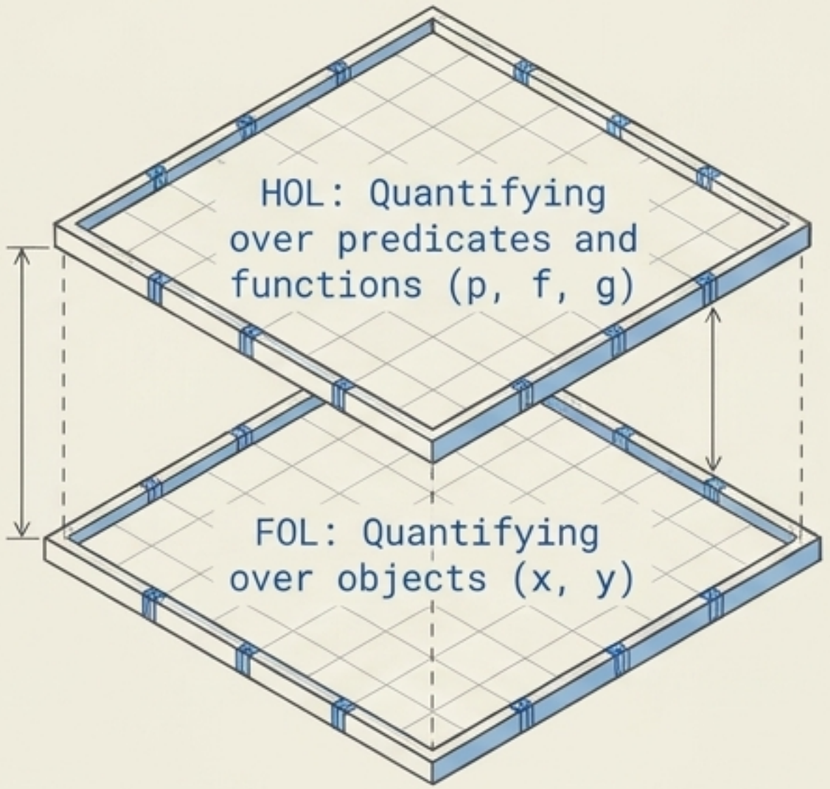
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The formula
$\forall f \forall g [(f=g) \leftrightarrow (\forall x f(x) = g(x))]$
states that two functions are equal if they have the same value for all possible arguments.
Why is this NOT a First-Order Logic statement?

A) It uses the biconditional (↔) operator.
B) It contains nested universal quantifiers.
C) It allows quantifiers ($\forall f$, $\forall g$) to range over functions, not just objects.
D) First-Order logic does not allow the equality (=) symbol.

Correct Answer: C

Rule: FOL only quantifies over objects. Second-order logic lifts this restriction, making it far more expressive, but entirely undecidable.



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Premise: The barber gives a haircut to a person if and only if that person doesn’t give a haircut to himself.

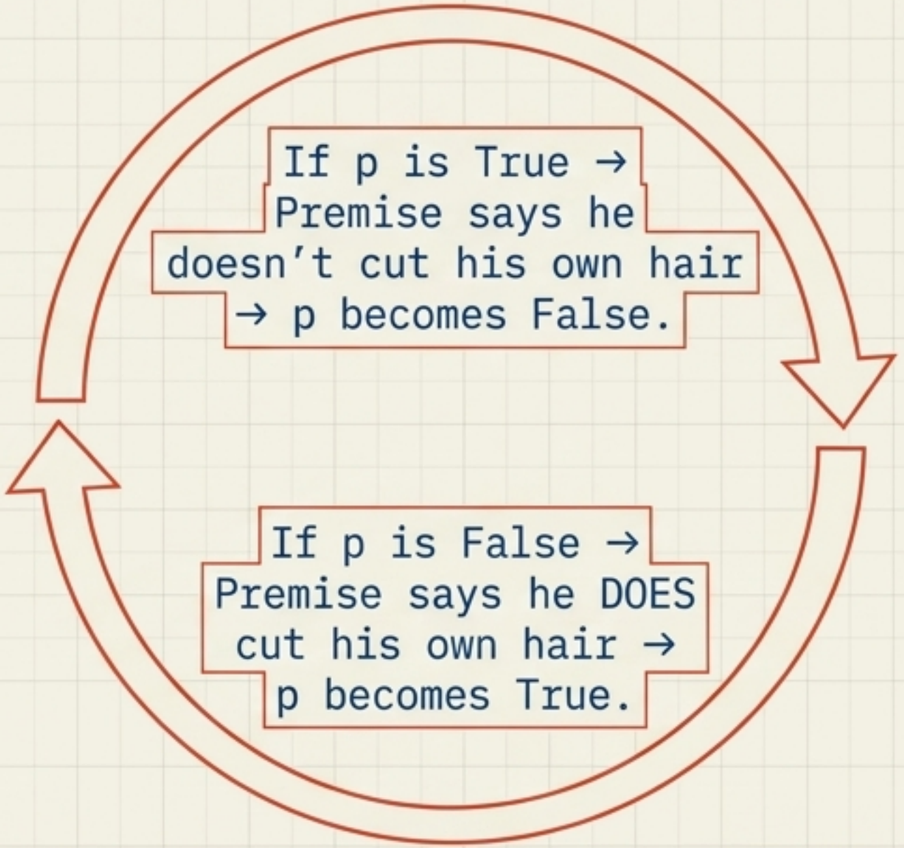
Let p be: The barber gives a haircut to himself.

What is the truth value of p?

A) $p \equiv T$
B) $p \equiv F$
C) It is a dilemma; p cannot be proven True or False.
D) $p \equiv T \wedge F$

Correct Answer: C

Conclusion: We cannot determine a consistent truth value.



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According to Gödel's Incompleteness Theorem (1931), which of the following is true for any logic inference system?

A) It can be made both perfectly consistent and perfectly complete by adding new axioms.

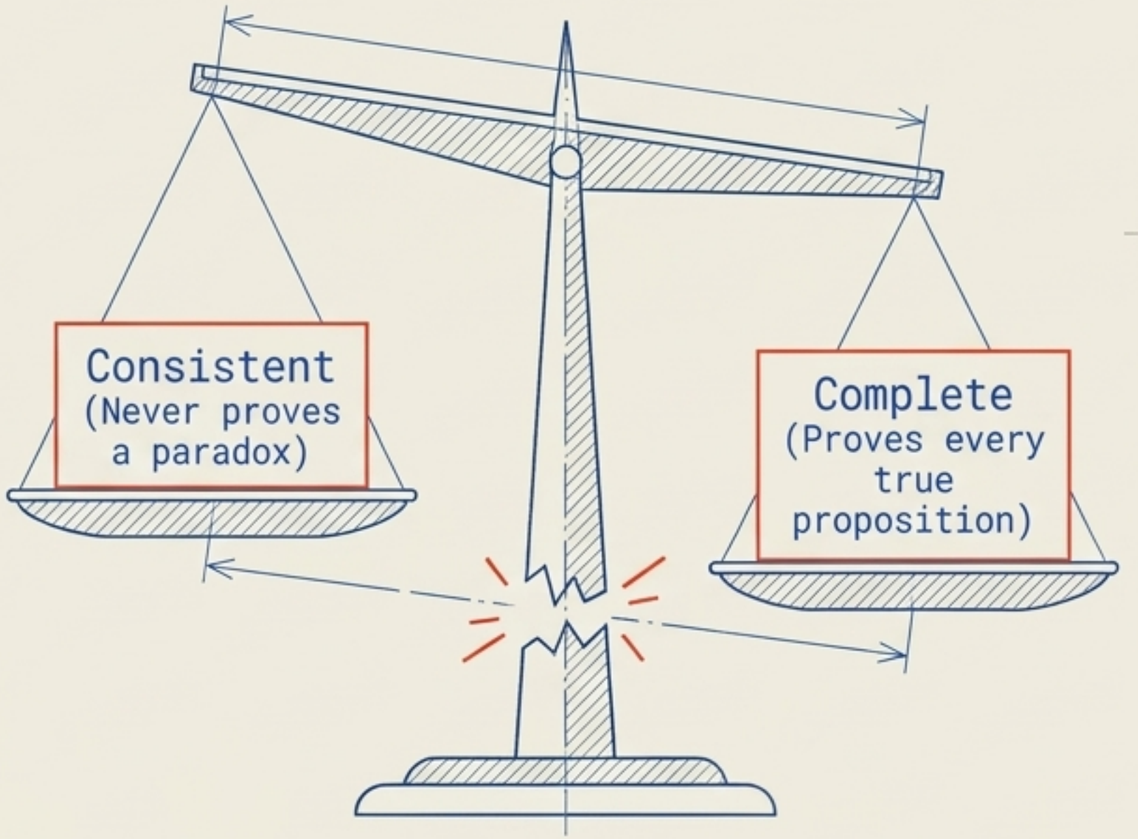
B) Any complete logic inference system must be decidable.

C) Any consistent logic inference system MUST BE incomplete.

D) All inconsistent systems are complete.

Correct Answer: C

Rule: If a system is consistent, there are inherently true statements that it simply cannot prove.



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The Architecture Mastered

- Syntax: Strict rules dictate WFFs.
- Mechanics: Quantifier order and scope fundamentally alter truth.
- Semantics: Interpretations map syntax to the universe.
- Limits: Unification Occurs Checks to Gödel's Incompleteness.



- Real-World Application: Vector Search and RAG Systems
- Guest Speaker: Dr. Alastair McKinley (Head of AI, SciLeads)
- Date: Wed, Oct 23, 2024 | 1:30pm-2:30pm
- Venue: BC-03-123A, Belfast Campus

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